NRES 746 – Workout of the Day #2: Pseudocoding a Permutation Test

Name: ____

Instructions: Work alone for about 5 minutes on Parts 1-2. Then get together with your neighbor(s) and spend about 5 minutes on Part 3, translating your pseudocode into actual R syntax **by hand, on paper** (no laptops).

Background: what is a permutation test?

In previous stats classes and in Lecture 1, you've seen tests for whether a sample's mean differed from a known null distribution: e.g., a z-test, and a t-test. These tests rely on knowing (or assuming) the shape of the population distribution the data came from.

A **permutation test** drops that requirement. It's a *nonparametric* (distribution-free) test: instead of simulating data from an assumed distribution, we generate a null distribution for our test statistic by **randomly reshuffling the labels** (e.g., treatment vs. control) on the data we already have, over and over. If the group labels really don't matter, shuffling them shouldn't change/reduce the expected difference between group means very much. If the observed difference is much larger than what typical reshufflings produce, that's evidence the labels *do* actually matter.

Part 1: The scenario

Recall the pygmy short-horned lizard example from Lecture 1: we want to know whether body mass differs between lizards in a cheatgrass-removal treatment (trtA) and a control group (Control).

```
trtA:      175  168  168  190  156  181  182  175  174  179
Control:   185  169  173  173  188  186  175  174  179  180
```

1a. What is the observed test statistic here (the single number we'll compare against the null distribution)?

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1b. Is the alternative hypothesis one-tailed or two-tailed in this case? Briefly explain why.

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Part 2: Write the pseudocode

2. Without looking back at your notes, write out – in plain English, step by step – the algorithm you would need to run a permutation test on this data. Think about: what gets shuffled, what gets recomputed each time, how many times you repeat it, and how you turn the result into a p-value.

Part 3: Translate to R (group, pen and paper)

3. As a group, fill in the blanks below to turn your pseudocode into real R syntax. You don't need working code – just your best attempt at the correct functions and syntax, written by hand.

```
permutation_test <- function(data, group, value, reps){

  observed_dif <- ___________________________________________

  null_difs <- ___________________________________________

  for(i in 1:reps){

    shuffled_labels <- ___________________________________________

    null_difs[i] <- ___________________________________________

  }

  p_value <- ___________________________________________

  return(p_value)
}
```

3a. Which R function did your group use to do the actual shuffling?

3b. What was the trickiest part of going from pseudocode to R syntax? Flag it – we'll come back to it!